

(SSL) Self Supervised Learning

Self Supervision

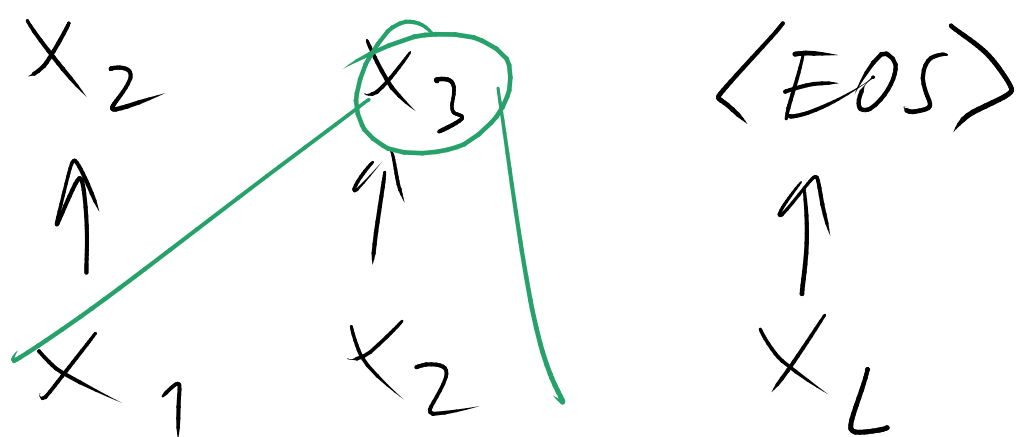
$$\{(x_i, y_i)\}_{i=1}^L \quad \frac{1}{L} \sum_{i=1}^L L(a(x_i), y_i) \quad \dots$$

↑
supervised

$$\{x_i\}_{i=1}^L$$

- Word2Vec, FastText
 - Autoencoders
 - Contrastive Learning
 - Language Models
-

Text



GPT

Architecture - Transformer Decoder

Task: Language modeling $p(x_1) p(x_2 | x_1)$

BERT (Bidirectional Encoder Representations from

Transformers)

Architecture - Transformer Encoder

Masked Language Modeling

| Shark | is | a | kind | of | fish | . | ← output

⇓ creation
of sample

↑ Transformer (target)

| [MASK] | is | a | kind | of | [MASK] | . | ← input

⇓

| Tralalero | is | a | kind | of | brainrot | .

How to mask?

All token

15%

85%

masked

unmasked



80% 10% 10%

[Mask] random token unchanged

$$\text{BERT loss} = L_{\text{MLM}} + L_{\text{NSP}}$$

NSP - Next sentence prediction

Model input:

$[CLS]_{256} t_1 t_2 \cancel{t_3} \dots t_N [SEP] s_1 s_2 \dots \cancel{s_{M+1}} s_M$
HEAD¹ [Mask] [Mask]

0 - s is not the next sentence after t

1 - s is the next sentence after t

$$\begin{aligned}
 & E_{[CLS]} E_{t_1} E_{t_2} E_{\dots} E_{t_N} E_{[SEP]} E_{s_1} \dots E_{[MASK]} E_{s_M} \\
 & + \\
 & E_0 E_1 E_2 \dots E_N E_{N+1} E_{N+2} \dots E_{N+M+1} \text{ (pos)} \\
 & + \\
 & E_A E_A E_A \dots E_A E_A E_B \dots E_B
 \end{aligned}$$

$SEP / \text{masked} = \text{nn.Embed}()$

Downstream BERT tasks?

GLUE (General Language Understanding Evaluation)

9 NLP tasks

— CoLA corpus of language acceptability
is sent. grammar ok? 2cls

— MNLI multi-genre Natural

3cls

Language Interference

9 — all classification

BERT 8/9 tasks

• NLP

• Images

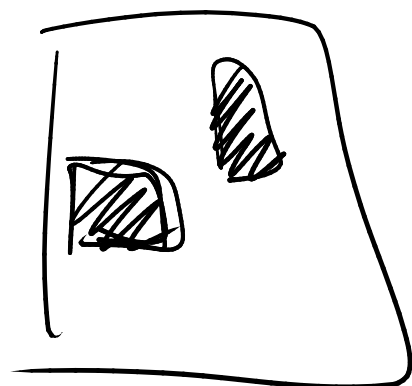
• Audio

ROBERTA

BEiT

HUBERT

ALBERT



SSL for Images

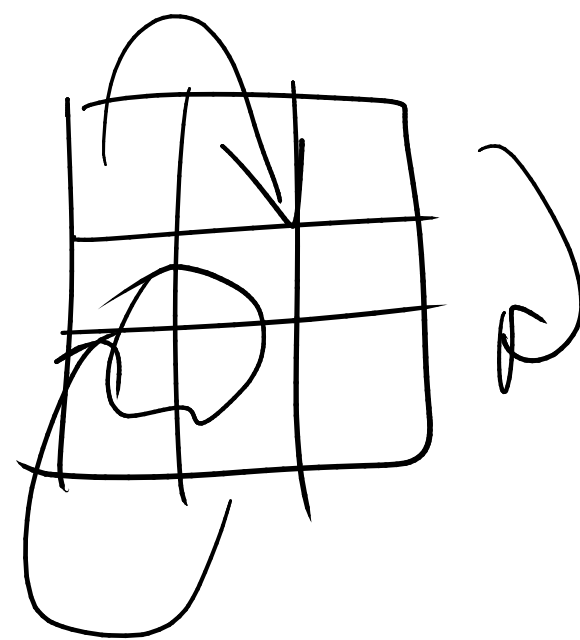
ImageNet is too good!

Pre-text tasks:

— Rotation angle prediction

90° , 180°

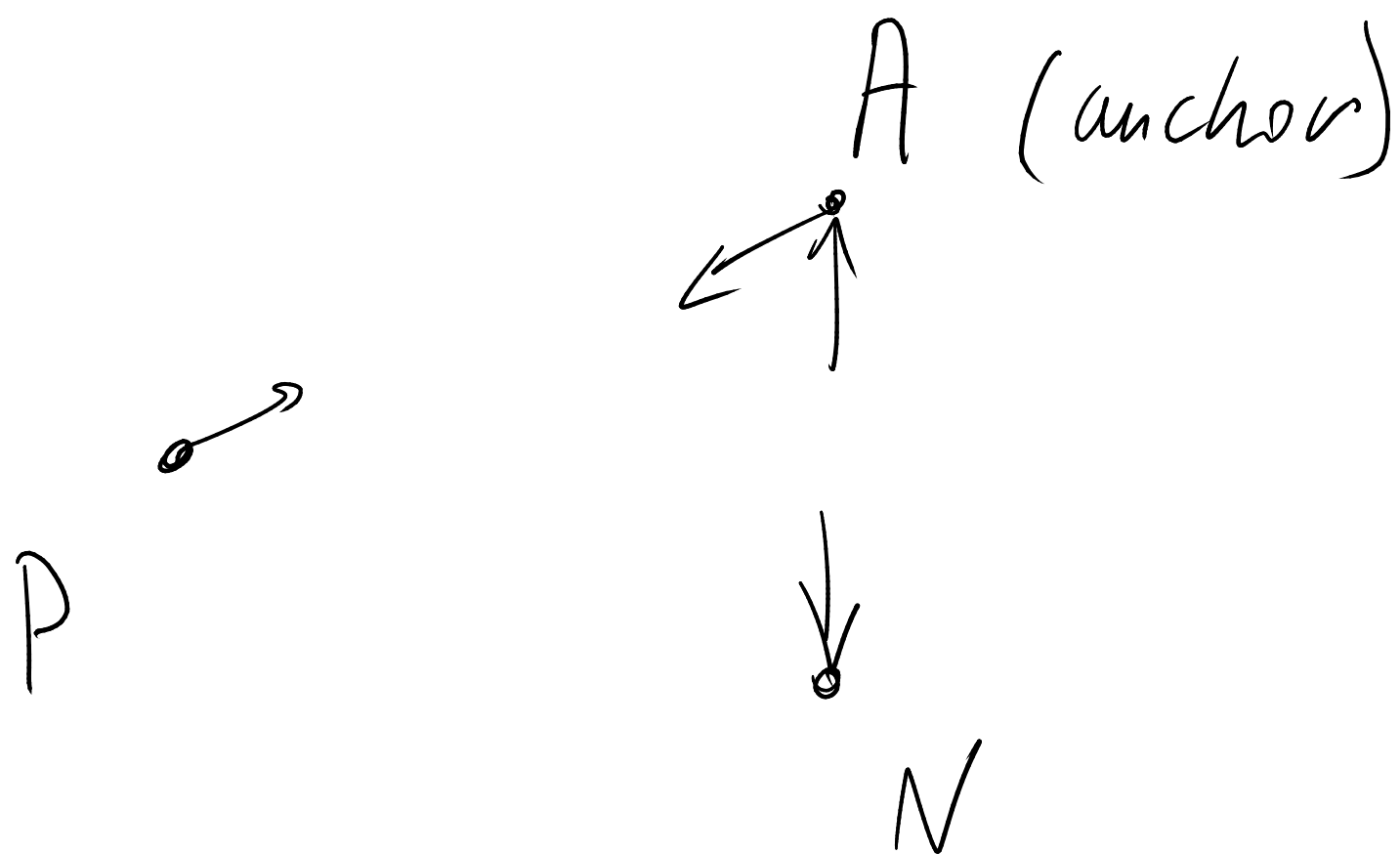
— Jigsaw puzzle



— Image colorization
grayscale $\rightarrow$ color

Triplet loss

Face Net $\|f(x) - f(y)\|_2^2$



$$\max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + m, 0)$$

$$m > 0$$

$$\downarrow$$

$$\min_f$$

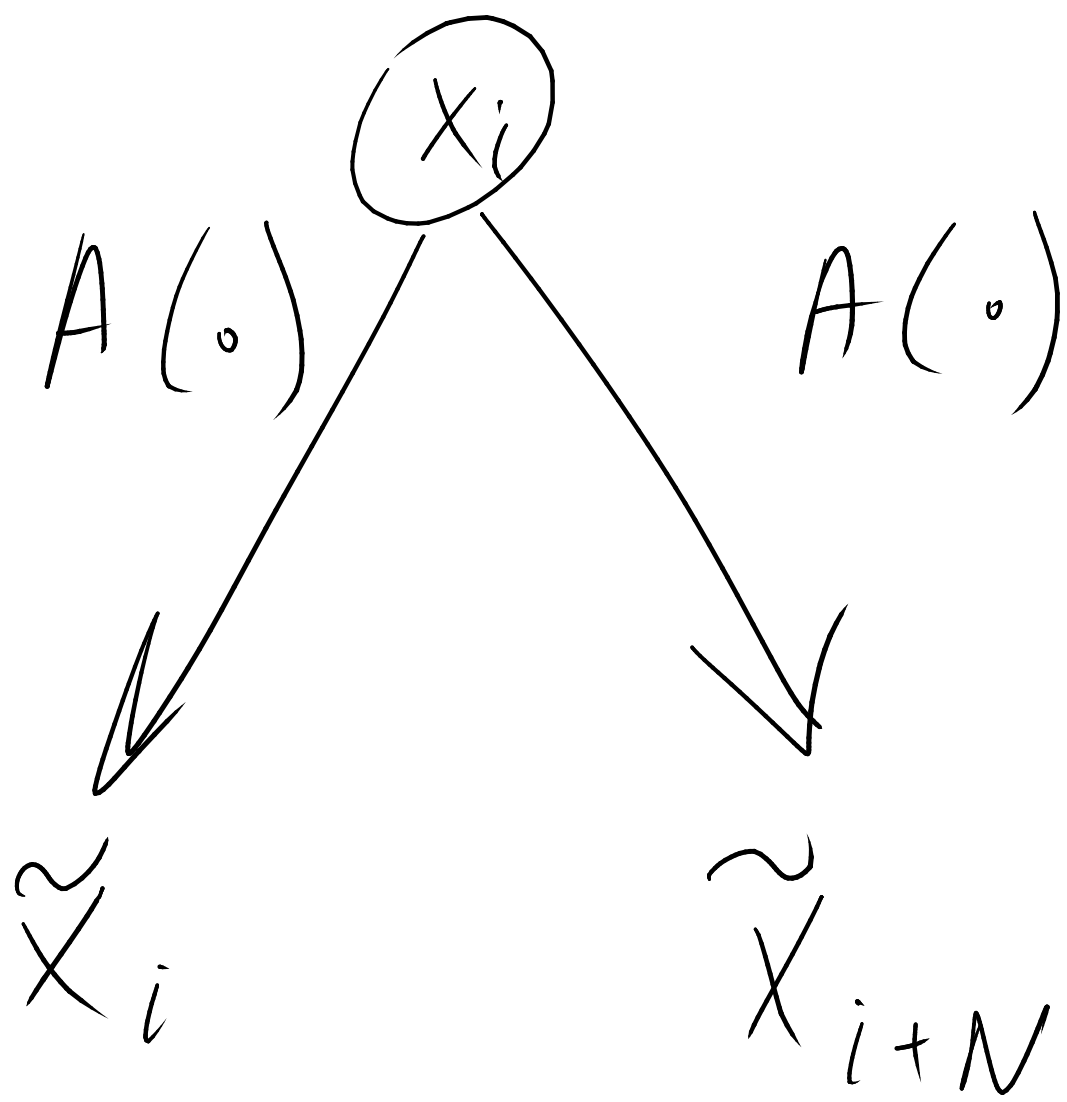
Contrastive Learning

SimCLR (A Simple Framework for Contrastive Learning of Representations)

$\{x_i\}_{i=1}^L$ Sample
 $\{x_i\}_{i=1}^N \leftarrow \text{batch}$
 $\sim A(-) \sim A'(x_1, \dots, x_N)$
 $\underbrace{x_1 \dots x_N}_{N} \quad \underbrace{\tilde{x}_{N+1} \dots \tilde{x}_{2N}}_{N}$

$$\{\tilde{x}_i\}_{i=1}^{2N}$$

double
batch of
augmented



A-random

$f \downarrow$

t_i

$g \downarrow$

z_i

$f \downarrow$

t_{i+N}

$g \downarrow$

$\leftarrow \text{head}$

z_{i+N}

maximize
agreement

$$g(\underbrace{f(x)}) = t$$

$\parallel$
 t

drop
before
fine
tuning

z_i, z_{i+N} - close

z_i, z_j $j \neq i+N$ $j \in [1, 2N]$

- far

$$\ell(z_i, z_{i+N}) = -\log \frac{\exp(\text{sim}(z_i, z_{i+N}))}{\sum_{k=1}^{2N} \mathbb{1}\{k \neq i, i+N\} \exp(\text{sim}(z_i, z_k))}$$

$\downarrow$
 $\min_{f, g}$

$$(A(x), x)$$

$$(A(x), A'(x))$$

- Random Crop
- Color Jitter
- Grayscale
- Gaussian Blur